

A Pole Hierarchy for Valley-Monotone Compositions

Resolving a bargraph asymptotic conjecture linked to OEIS A001523,
with exponential expansions, limit laws, and Lambert- W inversion

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Abstract

A composition is valley-monotone when the heights of its interior valley plateaux are weakly increasing from left to right. These are the non-decreasing bargraphs of Flórez, Ramírez, and Villamizar (2024). Their area-counting sequence was conjectured to have purely exponential growth, on the basis of a ratio at area 100. We give a self-contained analytic proof of that growth law, exact definitions of its constants, and integer-arithmetic certificates for their numerical values. The growth constant is determined directly by the generating function of the classical weakly unimodal composition sequence OEIS A001523.

The proof reveals a stronger structure. A decomposition by minimum part produces infinitely many genuine, simple positive poles, increasing to one, whose coefficient contributions alternate in sign. Consequently the area generating function is not D-finite and its coefficients are not P-recursive. We prove a finite-pole expansion to every prescribed radius below one, a central limit theorem for the number of bottom-level valleys, and an exponentially tight limiting terminal component. The accumulating real poles admit an all-orders inverse-logarithmic expansion expressed through Lambert W , with an error beyond every inverse-logarithmic order. We also give exponential-sector inversion and the necessary integer threshold qualifications. The claims are conventional mathematical proofs, accompanied by reproducible checks, not Lean formalizations or independently peer-reviewed results.

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1 The question, the OEIS connection, and the scope of the result

1.1 A published asymptotic conjecture

Let $d(n)$ count nonempty compositions of n whose valley heights are weakly increasing. A valley is a maximal constant run with larger parts on both sides; endpoint runs are not valleys. For example, $(3, 1, 2, 2, 4, 2, 3)$ has valley heights 1, 2, whereas $(3, 2, 3, 1, 2)$ has heights 2, 1 and is excluded. The area of the corresponding bargraph is the sum of its column heights. The sequence begins

$$1, 2, 4, 8, 16, 32, 64, 128, 256, 512, 1023, 2042, 4071, 8106, 16121, \dots$$

The convention throughout is $d(0) = 0$.

Conjecture 2.3 of Flórez, Ramírez, and Villamizar [1] proposes an exponential equivalent for this sequence. The displayed numerical estimates in that conjecture are

$$\lambda_{100} = 1.9764206780767492582\dots, \quad C_{100} = 0.6039015148966519\dots \quad (1.1)$$

Here the growth estimate is the finite ratio $d(100)/d(99)$, not an exact definition of the limiting exponential base. We establish

$$d(n) = C\lambda^n(1 + O(\theta^n)), \quad 0 < \theta < 1, \quad (1.2)$$

with

$$\begin{aligned} \lambda &= 1.97642067794583573531465245217332\dots, \\ C &= 0.60390151906189910496030721536293\dots \end{aligned}$$

Thus the proposed *form* is correct, while some displayed numerical digits must be replaced. Interpreting a finite decimal base literally in an asymptotic equivalent would be incorrect: an arbitrarily small fixed error in that base becomes exponentially significant as n grows.

1.2 What is, and is not, being claimed about OEIS

Let

$$A(q) = \sum_{n \geq 0} a(n)q^n, \quad a(n) = \text{OEIS A001523}(n). \quad (1.3)$$

This is the classical sequence of weakly unimodal compositions, with $a(0) = 1$. Its leading stretched-exponential asymptotic is already known; the OEIS entry credits Auluck (1951) and links the 2024 bargraph paper [2, 1]. We do *not* claim to solve a previously open leading-asymptotic problem for A001523 itself.

Instead, the classical sequence is the exact input to a new pole analysis of the associated valley-monotone class. In particular, the reciprocal growth constant $\rho = \lambda^{-1}$ is the unique solution in $(0, 1)$ of

$$\boxed{\rho(1 - \rho)A(\rho) = 1 + \rho.} \quad (1.4)$$

No OEIS identifier for $d(n)$ was verified in the searches for this report, so none is invented. The connection to OEIS is both explicit and structural, rather than an accidental match of a few terms.

1.3 Relationship to ProveIt and novelty boundary

The repository’s **Analysis/Transseries** project separates asymptotic scales, flatness, and inverse expansions from the Fabius function development [4]. The present article fits that program, but does not require any unproved theorem from the repository. A repository search for **bargraphs** and A001523 returned no matching files. This was a search-level overlap check, not an audit of every archive in the repository. The inspected repository tree had SHA 29aca108ed25d714f31b1316512777fbbdc8e006.

The generating-function decomposition is due to the 2024 paper; it is reproved below in minimum-part notation. The contributions developed here are the analytic resolution of the conjectured exponential form, the full positive-pole hierarchy and non-D-finiteness, the pole accumulation expansion, and the probabilistic and inverse consequences. Targeted searches located no earlier proof of this particular conjecture or these stronger conclusions. That is a bounded literature-search statement, not a guarantee of priority. General meromorphic coefficient extraction and analytic inversion are standard tools [3].

[write] **Added 2 October 2026, batch 75.** This report is a single manuscript, printed in full: manuscript 04 of batch 75 of ProveIt’s incoming-reports intake, delivered as the archive **OEIS_Valley_Monotone_Bargraphs.zip** in commit 4b874cea0 and placed in commit 6ea60e367 in the research-report collection, among the enumerative-combinatorics reports (**SetTheory/Cardinals/docs/reports**), not in the **Analysis/Transseries** project named above. The identifier called a “tree SHA” above is the ProveIt *commit* 29aca108e (1 October 2026), an ancestor of the placement commit; it is this manuscript’s pin. Nothing in this report is formalized in Lean or Rocq, and placement in the collection confers no formal status on any statement here. No OEIS number for the sequence $d(n)$ was verified, and the writing step assigns none; A001523 is the input sequence, not the target. The writing step changed the labels (every label now carries the prefix **vmb:**), renamed Theorem 7.3 (delivered as “Pole accumulation transseries”), set the bibliography ragged-right, and added these [write] notes and one reference [5]; no other statement, proof or number of the manuscript was altered. The Lambert- W and rounding steps of Sections 7 and 8 are instances of the transseries-and-inversion volume [5], as the notes there say; no transseries in that volume’s sense is constructed here.

2 A decomposition by minimum part

2.1 Classes and notation

For an integer $j \geq 1$, let $\mathcal{U}_j$ be the nonempty weakly unimodal compositions whose parts are at least j , and let $\mathcal{D}_j$ be the nonempty valley-monotone compositions with that same lower bound. Write

$$U_j(q) = \sum_{\alpha \in \mathcal{U}_j} q^{|\alpha|}, \quad D_j(q) = \sum_{\alpha \in \mathcal{D}_j} q^{|\alpha|}, \quad d_j(n) = [q^n]D_j(q). \quad (2.1)$$

Here $|\alpha|$ is the sum of the parts. In particular, $U_1 = A - 1$, $D_1(q) = \sum_{n \geq 1} d(n)q^n$. Define

$$H_j(q) = \frac{q^j}{1 - q^j}, \quad K_j(q) = H_j(q)U_{j+1}(q), \quad Q_j(q) = 1 - q^j - q^j U_{j+1}(q). \quad (2.2)$$

Thus

$$Q_j(q) = (1 - q^j)(1 - K_j(q)). \quad (2.3)$$

The series H_j counts a nonempty run of parts equal to j .

Lemma 2.1 (Unimodal decomposition). *As formal power series,*

$$U_j = H_j + \frac{U_{j+1}}{(1 - q^j)^2}. \quad (2.4)$$

Equivalently,

$$U_j(q) = \sum_{h \geq j} \frac{q^h}{1 - q^h} \prod_{r=j}^{h-1} (1 - q^r)^{-2}. \quad (2.5)$$

Proof. An object either consists only of parts j , or has a nonempty middle object with all parts greater than j . In the second case, unimodality allows parts j only in the initial and final runs. Those runs may be empty and are chosen independently, giving $(1 - q^j)^{-2}$. This proves the recurrence. Iteration classifies an object by its maximum part h and gives the sum. For each coefficient of fixed degree, only finitely many values of h contribute, so the formal iteration is legitimate. $\square$

Lemma 2.2 (Valley-monotone decomposition). *As formal power series,*

$$D_j = H_j + \frac{D_{j+1}}{(1 - q^j)^2(1 - K_j)} = H_j + \frac{D_{j+1}}{(1 - q^j)Q_j}. \quad (2.6)$$

Proof. The all- j object contributes H_j . Otherwise split the composition into its maximal blocks of parts greater than j , with runs of j 's between them and possibly at the two ends. Each internal separating run is a valley of height j .

Every block except the last must be unimodal. Indeed, an internal valley in an earlier block would have height greater than j and would be followed by a separating valley of height j , violating the required order. A finite sequence without an internal valley is weakly unimodal: a downward change followed later by an upward change necessarily creates an intervening valley. Conversely, unimodal earlier blocks create no additional valleys and the last block may be any member of $\mathcal{D}_{j+1}$.

The unique decomposition is therefore

$$(\text{initial } j\text{-run}) (\mathcal{U}_{j+1} \text{ nonempty } j\text{-run})^* \mathcal{D}_{j+1} (\text{final } j\text{-run}).$$

The two endpoint factors contribute $(1 - q^j)^{-2}$; the repeated unit has generating function K_j . This proves the formula, including uniqueness and the absence of overcounting. $\square$

Corollary 2.3 (Iterated series). *For $j \geq 1$,*

$$D_j(q) = \sum_{h \geq j} H_h(q) \prod_{r=j}^{h-1} \frac{1}{(1 - q^r)Q_r(q)}. \quad (2.7)$$

In particular,

$$D_1(q) = \sum_{h \geq 1} \frac{q^h}{\prod_{r=1}^h (1 - q^r) \prod_{r=1}^{h-1} Q_r(q)}. \quad (2.8)$$

This agrees with the area specialization in [1], where the paper's B_j is our U_{j+1} . The boundary condition at large j is coefficientwise disappearance: U_j, D_j have no terms below degree j . It uniquely determines both recurrences.

2.2 The renewal kernel in terms of A001523

From (2.4) at $j = 1$,

$$U_2 = (1 - q)^2 U_1 - q(1 - q).$$

Consequently,

$$K_1(q) = q(1 - q)A(q) - q = \sum_{n \geq 3} (a(n - 1) - a(n - 2))q^n. \quad (2.9)$$

Thus the differences of a classical OEIS sequence count the repeated units. This identity will also provide the probability distribution underlying the limit theorem.

3 Analytic continuation and the positive-pole hierarchy

3.1 Normal convergence inside the unit disk

Lemma 3.1 (Uniform estimates). *Fix $0 < R < 1$. Uniformly for $|q| \leq R$, all the functions U_j are analytic and*

$$|U_j(q)| \leq B_R R^j, \quad |K_j(q)| = O_R(R^{2j+1}), \quad Q_j(q) = 1 + O_R(R^j), \quad (3.1)$$

where B_R is finite and independent of j .

Proof. The infinite product $P_R = \prod_{r \geq 1} (1 - R^r)^{-2}$ is finite, since the sum of the absolute logarithms converges. Each term in (2.5) is bounded by $P_R R^h / (1 - R)$. Summing for $h \geq j$ gives $B_R = P_R / (1 - R)^2$, and local uniform convergence proves analyticity. The other estimates follow from (2.2). $\square$

Theorem 3.2 (Meromorphic continuation). *Every D_j extends to a single-valued meromorphic function on $|q| < 1$. Its possible poles there are among the zeros of Q_r , $r \geq j$. On each compact subdisk there are only finitely many possible poles.*

Proof. For sufficiently large r , uniformly on a chosen compact subdisk, Q_r stays away from zero and

$$\frac{1}{(1 - q^r)Q_r(q)} = 1 + O_R(R^r).$$

The corresponding tail products in (2.7) are uniformly bounded, because $\sum_r R^r$ converges. The summands are therefore $O_R(R^h)$ after separating finitely many initial denominator factors. The tail sum converges normally off those factors' zeros. Multiplying by the finite product of initial Q_r 's makes the sum analytic locally. Each initial Q_r is analytic and not identically zero, so its zeros are isolated and finite in a compact subdisk. These observations prove all the assertions. $\square$

3.2 Distinguished real roots

Lemma 3.3. *For every $j \geq 1$ there is a unique $\rho_j \in (0, 1)$ with $K_j(\rho_j) = 1$, equivalently $Q_j(\rho_j) = 0$. Moreover,*

$$0 < \rho_1 < \rho_2 < \cdots < 1, \quad \rho_j \longrightarrow 1, \quad Q'_j(\rho_j) < 0. \quad (3.2)$$

The only zero of Q_j on $|q| \leq \rho_j$ is $q = \rho_j$.

Proof. On $(0, 1)$, K_j is strictly increasing, starts at zero, and tends to infinity. The last claim follows already from the subclass consisting of one part greater than j , together with the nonempty separating run. Thus ρ_j exists uniquely and its derivative is positive. Equation (2.3) then gives $Q'_j(\rho_j) = -(1 - \rho_j^j)K'_j(\rho_j) < 0$.

For positive $q < 1$, both $H_j > H_{j+1}$ and $U_{j+1} > U_{j+2}$. Hence $K_j > K_{j+1}$, proving strict ordering of their unit crossings. The estimates in Lemma 3.1 show that $K_j(q) \rightarrow 0$ for every fixed $q < 1$; this forces $\rho_j \rightarrow 1$.

If $|q| < \rho_j$, positivity gives $|K_j(q)| \leq K_j(|q|) < 1$, excluding zeros. On the boundary, equality with $K_j(q) = 1$ in the triangle inequality would force all nonzero summands of $K_j(q)$ to be nonnegative real. Its support contains both $2j + 1$ and $2j + 2$: use a separating part j and a singleton of height $j + 1$ or $j + 2$. Their consecutive exponents force $q/\rho_j = 1$. This proves boundary uniqueness. $\square$

Theorem 3.4 (Genuine simple poles and alternating amplitudes). *For every $k \geq j$, D_j has a genuine simple pole at ρ_k . In particular, write the principal part of D_1 there as*

$$\frac{C_k}{1 - q/\rho_k}.$$

Then

$$C_k = \frac{D_{k+1}(\rho_k)}{(1 - \rho_k^k)(-\rho_k Q'_k(\rho_k))} \prod_{r=1}^{k-1} \frac{1}{(1 - \rho_k^r)Q_r(\rho_k)}, \quad \text{sgn}(C_k) = (-1)^{k-1}. \quad (3.3)$$

These are exactly the positive real poles inside the unit disk.

Proof. By Lemma 3.3, D_{k+1} is analytic at ρ_k : all of its possible pole factors have indices greater than k and larger minimum zero modulus. Its defining positive series converges there and has positive value. Applying (2.6) at level k therefore produces a genuine simple pole, of amplitude

$$\frac{D_{k+1}(\rho_k)}{(1 - \rho_k^k)(-\rho_k Q'_k(\rho_k))} > 0.$$

For every $r < k$, the recurrence at level r adds an analytic term and multiplies the inherited pole by a finite nonzero factor $((1 - q^r)Q_r(q))^{-1}$. Since $Q_r(\rho_k) < 0$, each such factor changes its sign. This proves the formula, noncancellation, and sign. Theorem 3.2 and uniqueness of each positive zero exclude any other positive poles. $\square$

Corollary 3.5 (Non-D-finiteness and non-P-recursiveness). *None of the functions D_j is D-finite over $\mathbb{C}(q)$. In particular $d(n)$ satisfies no eventual nonzero linear recurrence with polynomial coefficients in n .*

Proof. A function satisfying a linear differential equation with polynomial coefficients can have finite-plane singularities only at the finitely many zeros of its leading coefficient. At every other point, standard local existence for a linear analytic differential equation continues its analytic branches holomorphically. The infinitely many poles ρ_k , $k \geq j$, contradict this requirement.

For completeness, a polynomial-coefficient recurrence for the Taylor coefficients, after multiplication by powers of q and summation, yields a linear differential equation for the generating function, using $q d/dq$ to encode multiplication by n . Any finite initial discrepancy is polynomial and may be annihilated by further differentiation. Thus an eventual recurrence would also imply D-finiteness, which has just been excluded. $\square$

4 The exponential law and certified constants

Theorem 4.1 (Leading asymptotics at every fixed minimum part). *For each fixed $j \geq 1$, there is $R_j > \rho_j$, with $R_j < 1$, such that*

$$d_j(n) = c_j \rho_j^{-n} + O_j(R_j^{-n}), \quad c_j = \frac{D_{j+1}(\rho_j)}{(1 - \rho_j^j)^2 \rho_j K'_j(\rho_j)} > 0. \quad (4.1)$$

For $j = 1$, $c_1 = C_1 = C$ and $\rho_1 = \rho$, so this proves (1.2) and the intended form of Conjecture 2.3 of [1].

Proof. Lemma 3.3 and Theorem 3.4 give a unique simple pole on the circle of convergence. Local finiteness of all possible poles permits a slightly larger closed disk containing no others. Subtract the principal part $c_j/(1 - q/\rho_j)$. The difference is analytic on a neighborhood of that disk. Cauchy's coefficient bound on its boundary gives the stated error. The formula for c_j follows from (2.3). Equation (2.9) gives (1.4). $\square$

There are no nonzero inverse-power corrections multiplying the leading exponential: for every fixed M ,

$$\frac{d(n)}{C\lambda^n} = 1 + o(n^{-M}). \quad (4.2)$$

The missing information is instead in genuinely smaller exponential sectors. This is why an inverse-power fit to the leading term would conceal the analytic structure.

4.1 Exact definitions and integer-arithmetic enclosures

The constants are defined by

$$K_1(\rho) = 1, \quad \lambda = \rho^{-1}, \quad C = \frac{D_2(\rho)}{(1 - \rho)^2 \rho K'_1(\rho)}. \quad (4.3)$$

The accompanying `verify.py` proves the following enclosures with outward-rounded rational intervals, using integers only:

$$\begin{aligned} 0.50596515770080666537292224451833314155847753935777 &< \rho, \\ \rho &< 0.50596515770080666537292224451833314155847753935778. \end{aligned} \quad (4.4)$$

$$1.976420677945835735314652452172 < \lambda < 1.976420677945835735314652452175, \quad (4.5)$$

$$0.603901519061899104960307215362 < C < 0.603901519061899104960307215364. \quad (4.6)$$

The narrower machine-produced intervals are retained in `verification.json`. These bounds certify, in particular, the numerical distinction between the limit and the finite-area estimates.

Proposition 4.2 (Tail bounds used in the certificate). *For $0 < q < 1$, put $S_J(q) = q^{J+1}/(1-q)$. If $S_J(q) < 1$, then*

$$0 \leq U_{J+1}(q), D_{J+1}(q) \leq \frac{S_J(q)}{1 - S_J(q)}, \quad 0 \leq U'_{J+1}(q) \leq \frac{S'_J(q)}{(1 - S_J(q))^2}. \quad (4.7)$$

Proof. S_J counts a single allowed part. All nonempty ordered sequences of such parts have generating function $S_J/(1 - S_J)$, and contain both restricted classes as subclasses. Coefficientwise domination proves the first two inequalities and, after differentiating nonnegative coefficients, the derivative bound. $\square$

The program uses $J = 250$ and endpoints in $10^{-90}\mathbb{Z}$, propagates these tail intervals down to level 2, and tests the sign of Q_1 at the two rational endpoints in (4.4). To enclose C , it uses the equivalent expression

$$C = \frac{D_2(\rho)}{\rho(1 - \rho)\{1 + U_2(\rho) + \rho U'_2(\rho)\}}. \quad (4.8)$$

Every division checks that its denominator interval excludes zero. No unbounded numerical tail is discarded.

5 Exponential expansions to arbitrary subunit radius

Theorem 5.1 (Finite-pole expansion). *Choose $0 < R < 1$ such that D_1 has no pole on $|q| = R$, and let $\mathcal{P}_R$ be its finite set of actual poles inside that circle. Write the principal part at $\zeta \in \mathcal{P}_R$ in the form*

$$\sum_{\ell=1}^{m_\zeta} \frac{c_{\zeta,\ell}}{(1 - q/\zeta)^\ell}.$$

Then

$$d(n) = \sum_{\zeta \in \mathcal{P}_R} \sum_{\ell=1}^{m_\zeta} c_{\zeta,\ell} \binom{n+\ell-1}{\ell-1} \zeta^{-n} + O_R(R^{-n}). \quad (5.1)$$

Proof. Subtract all the displayed principal parts. The result is analytic in a neighborhood of the closed disk, so its coefficients are $O_R(R^{-n})$ by Cauchy's formula. The coefficient of each subtracted rational term is the displayed binomial coefficient times ζ^{-n} . $\square$

For positive real poles, $m_{\rho_k} = 1$ and $c_{\rho_k,1} = C_k$. Thus these sectors have no polynomial prefactors. Nonreal conjugate pairs may yield exponentially small oscillations; multiplicities or coincidences among complex zeros may yield polynomial factors. Formula (5.1) makes no unsupported simplicity assumption for those zeros.

The numerical positive-real hierarchy begins as follows. The table's ordering is by *positive real pole*, not by modulus of all complex poles. Apart from the first row, these decimal entries are numerical checks rather than rational certificates.

k	ρ_k	ρ_k^{-1}	C_k
1	0.505965157700807	1.976420677945836	0.603901519061899
2	0.628520593778803	1.591037763755333	-0.434266071916591
3	0.694915251028261	1.439024396889127	0.097127120171435
4	0.738032724347822	1.354953468877238	-0.008189474598318
5	0.768802380990748	1.300724379536014	0.000290088602322
6	0.792097830048713	1.262470318771736	-0.000004651415177

Remark 5.2 (Meaning of “all orders”). For every fixed exponential cutoff $R < 1$, only finitely many sectors are required and the remainder is proved. This does not assert that the infinite sum of all pole contributions converges, that the unit circle is a natural boundary, or that the positive-real poles alone constitute the complete expansion. In particular, we have not certified that ρ_2 is the second-smallest modulus among *all* poles. An all-orders algebraic expansion around the dominant term is trivial here; the finite-pole theorem is the substantive exponential refinement.

6 A renewal limit law and a tight terminal component

6.1 Bottom-level valleys

Choose uniformly from the $d(n)$ objects of area n , and let V_n be the number of valleys of height 1. Introduce a variable u marking these valleys. The decomposition in Lemma 2.2 gives

$$D(q, u) = H_1(q) + \frac{B(q)}{1 - uK(q)}, \quad B(q) = \frac{D_2(q)}{(1 - q)^2}, \quad K(q) = K_1(q). \quad (6.1)$$

The repeated unit contributes exactly one height-1 valley. Define a positive integer random variable Z by

$$\mathbb{P}(Z = s) = [q^s]K(q)\rho^s. \quad (6.2)$$

This is a probability distribution because $K(\rho) = 1$. It has exponential moments near zero and includes both $s = 3$ and $s = 4$ in its support. Put

$$M = \mathbb{E}Z = \rho K'(\rho), \quad V = \text{Var } Z = \rho^2 K''(\rho) + \rho K'(\rho) - M^2 > 0. \quad (6.3)$$

Theorem 6.1 (Gaussian bottom-valley count). *As $n \rightarrow \infty$,*

$$\frac{V_n - \mu n}{\sqrt{\sigma^2 n}} \implies \mathcal{N}(0, 1), \quad \mu = \frac{1}{M}, \quad \sigma^2 = \frac{V}{M^3}. \quad (6.4)$$

Moreover $\mathbb{E}V_n = \mu n + O(1)$ and $\text{Var } V_n = \sigma^2 n + O(1)$. Numerical evaluation gives

$$\mu = 0.1372487560663833542\dots, \quad \sigma^2 = 0.0368978521690088765\dots$$

Proof. The implicit function theorem gives an analytic root $r(u)$ near $u = 1$, satisfying $uK(r(u)) = 1$ and $r(1) = \rho$. Choose a closed disk slightly larger than ρ in which (6.1) at $u = 1$ has only its simple pole. Compactness and analytic perturbation preserve exactly this one nearby pole for complex u sufficiently close to 1. Subtracting it uniformly gives

$$\mathbb{E}[u^{V_n}] = \frac{C(u)}{C(1)} \left(\frac{\rho}{r(u)} \right)^n (1 + O(\vartheta^n)), \quad C(u) = \frac{B(r(u))}{ur(u)K'(r(u))}, \quad (6.5)$$

for some $\vartheta < 1$, locally uniformly in u .

Set $t(s) = \log r(e^s)$. Taking logarithms of the root equation gives $\log K(e^{t(s)}) = -s$. The first two derivatives of $\log K(e^t)$ at $t = \log \rho$ are respectively M and V . Therefore

$$t'(0) = -M^{-1}, \quad t''(0) = -VM^{-3}.$$

Taking logarithms in (6.5) and differentiating twice proves the mean and variance statements. Substituting $s = iz/\sqrt{\sigma^2 n}$, subtracting the centering term, and using the analytic Taylor expansion gives a limiting characteristic function $\exp(-z^2/2)$. This proves the asserted weak convergence. $\square$

6.2 Why almost all valleys are at height one

In the nonflat decomposition, let $T_n \in \mathcal{D}_2$ be the final block with parts greater than one, and let $L_n, R_n \geq 0$ be the lengths of the two endpoint runs of ones. The all-ones composition has probability $1/d(n)$. On this exceptional object, define $(L_n, R_n) = (0, 0)$ and let $T_n = \dagger$ be a cemetery object of area zero. This convention is asymptotically irrelevant.

Theorem 6.2 (Boltzmann terminal limit). *The triple (L_n, T_n, R_n) converges in distribution to independent components (L, T, R) , where*

$$\mathbb{P}(L = \ell) = \mathbb{P}(R = \ell) = (1 - \rho)\rho^\ell, \quad \mathbb{P}(T = \tau) = \frac{\rho^{|\tau|}}{D_2(\rho)} \quad (\tau \in \mathcal{D}_2). \quad (6.6)$$

The sizes of these components have uniformly bounded exponential moments for all sufficiently large n . The number of valleys above height one is therefore $O_{\mathbb{P}}(1)$, and the total number of valleys has the same central limit theorem as V_n .

Proof. For a fixed triple (ℓ, τ, r) , its contribution is $q^{\ell+|\tau|+r}/(1-K(q))$. Comparing its dominant residue with that of (6.1) at $u = 1$ gives limiting probability

$$\frac{\rho^{\ell+|\tau|+r}}{B(\rho)} = (1-\rho)^2 \rho^{\ell+r} \frac{\rho^{|\tau|}}{D_2(\rho)},$$

which factors as stated and sums to one.

To justify tightness and exponential moments, mark endpoint length by v and terminal area by w . The numerator becomes $D_2(wq)/(1-vq)^2$. Since D_2 is analytic past ρ , this numerator remains analytic on a common larger disk when real $v, w > 1$ are sufficiently close to 1. The same residue argument then bounds the marked expectation uniformly in n . All valleys above height one lie in T_n , and their number is at most its area. Their contribution divided by $\sqrt{n}$ tends to zero in probability, so Slutsky's theorem transfers the central limit law to the total valley count. $\square$

The limiting terminal area has mean

$$\mathbb{E}|T| = \frac{\rho D'_2(\rho)}{D_2(\rho)} = 6.2446300331882976716 \dots$$

This decimal is numerical, not one of the certified intervals above. The same arguments work at every fixed level j , with K_j , D_{j+1} , and runs of parts j . As an additional consequence of Theorem 4.1,

$$\mathbb{P}(\text{all parts are at least } 2) \sim \frac{c_2}{c_1} \left(\frac{\rho_1}{\rho_2} \right)^n. \quad (6.7)$$

Here $c_2 > 0$ is the *dominant amplitude* of D_2 , not the negative amplitude C_2 of the second positive pole of D_1 .

7 How the real poles accumulate at one

7.1 An exact product comparison

For $0 < q < 1$, define

$$P_{j+1}(q) = \prod_{h \geq j+1} (1 - q^h)^{-2}. \quad (7.1)$$

Lemma 7.1. *If $y = q^j$, then*

$$\frac{1 - yq}{2 - yq} (P_{j+1} - 1) \leq U_{j+1} \leq \frac{1}{2} (P_{j+1} - 1). \quad (7.2)$$

Proof. Let $A_h = \prod_{r=j+1}^h (1 - q^r)^{-2}$, with $A_j = 1$. The term of U_{j+1} with maximum part h is

$$\frac{q^h}{1 - q^h} A_{h-1} = \frac{1 - q^h}{2 - q^h} (A_h - A_{h-1}).$$

The multiplicative factor lies between $(1 - yq)/(2 - yq)$ and $1/2$. Sum the positive telescoping increments and pass to the limit. $\square$

Put

$$t_j = -\log \rho_j, \quad x_j = jt_j, \quad y_j = e^{-x_j} = \rho_j^j, \quad c = \log 2. \quad (7.3)$$

At the pole, the defining equation is $U_{j+1}(\rho_j) = (1 - y_j)/y_j$.

Lemma 7.2 (Pole-scale equation). *As $j \rightarrow \infty$, $x_j \rightarrow \infty$, $x_j \sim \log j$, and*

$$\frac{2je^{-x_j}}{x_j} = x_j + c + O\left(\frac{x_j^3}{j}\right). \quad (7.4)$$

Proof. We know $t_j \rightarrow 0$. We first show $y_j \rightarrow 0$. Otherwise some subsequence has $y_j \geq \delta > 0$. The singleton subclass gives $U_{j+1}(\rho_j) \geq \rho_j^{j+1}/(1 - \rho_j) \rightarrow \infty$, whereas $(1 - y_j)/y_j$ is bounded, a contradiction. Thus $x_j \rightarrow \infty$.

At the root, (7.2) gives

$$\log P_{j+1}(\rho_j) = x_j + c + O(y_j). \quad (7.5)$$

On the other hand, expanding the product logarithm absolutely gives, with $t = t_j$, $y = y_j$,

$$\begin{aligned} \log P_{j+1}(e^{-t}) &= 2 \sum_{k \geq 1} \frac{y^k}{k(e^{kt} - 1)} \\ &= \frac{2y}{t} + O\left(y + \frac{y^2}{t}\right). \end{aligned} \quad (7.6)$$

For the last estimate use $(e^t - 1)^{-1} = t^{-1} + O(1)$ for $k = 1$, and $(e^{kt} - 1)^{-1} \leq (kt)^{-1}$ for $k \geq 2$.

Since $y \rightarrow 0$, the error is $o(y/t)$, while the right side of (7.5) tends to infinity. It follows that $2y/t \sim x$, or $2je^{-x} \sim x^2$. Taking logarithms gives $x + 2\log x = \log(2j) + o(1)$, hence $x \sim \log j$ and $y \asymp x^2/j$. Substituting this estimate back into (7.6) and (7.5) gives (7.4). $\square$

7.2 A Lambert– W expansion to all logarithmic orders

Theorem 7.3 (Pole accumulation: a Lambert– W expansion). *Let $X_j > 0$ be the unique positive solution of*

$$e^{X_j} X_j (X_j + c) = 2j, \quad c = \log 2, \quad (7.7)$$

and set

$$w_j = 2W\left(\sqrt{j/2}\right), \quad (7.8)$$

where W is the positive real Lambert function. Then

$$x_j = X_j + O\left(\frac{(\log j)^2}{j}\right). \quad (7.9)$$

For every fixed $m \geq 1$,

$$x_j = w_j + \sum_{k=1}^m \frac{\alpha_k(c)}{w_j^k} + O_m(w_j^{-m-1}) + O\left(\frac{w_j^2}{j}\right), \quad (7.10)$$

where

$$\begin{aligned} \alpha_1(c) &= -c, \\ \alpha_2(c) &= \frac{c(c+4)}{2}, \\ \alpha_3(c) &= -\frac{c(c^2+6c+12)}{3}, \\ \alpha_4(c) &= \frac{c(3c+8)(c^2+6c+12)}{12}. \end{aligned} \quad (7.11)$$

The full series for $X_j - w_j$ is convergent for sufficiently large w_j ; the difference between x_j and this logarithmic sector is smaller than every fixed inverse power of $\log j$.

Proof. Let $F_j(x) = 2je^{-x}/x - x - c$. Its derivative is negative. Lemma 7.2 gives $F_j(x_j) = O(x_j^3/j)$, while $F_j(X_j) = 0$. First, $F'_j \leq -1$ gives $|x_j - X_j| = O(x_j^3/j) = o(1)$. Thus throughout the interval between these two points, $|F'_j| \asymp x_j$, because $2je^{-x_j}/x_j \sim x_j$. A second application of the mean value theorem improves the difference to $O(x_j^2/j)$, proving (7.9).

The definition of $w = w_j$ is equivalent to $e^w w^2 = 2j$. Write $X = w + \varepsilon$, $z = 1/w$. Taking logarithms of (7.7) and cancelling the equation for w gives

$$\varepsilon + 2 \log(1 + z\varepsilon) + \log \left(1 + \frac{cz}{1 + z\varepsilon} \right) = 0. \quad (7.12)$$

This is analytic near $(z, \varepsilon) = (0, 0)$, with derivative 1 with respect to ε . The analytic implicit function theorem supplies a convergent series $\varepsilon = \sum_{k \geq 1} \alpha_k(c) z^k$. Coefficient comparison gives (7.11), and gives every further coefficient recursively: substitute the already computed terms, and negate the coefficient of z^k to obtain α_k . The new coefficient occurs only in the first, linear term at that order. Combining this convergent expansion with (7.9) proves all claims. $\square$

In particular,

$$1 - \rho_j \sim \frac{\log j}{j}, \quad x_j = \log j - 2 \log \log j + \log 2 + o(1). \quad (7.13)$$

Higher expansions for $\rho_j = e^{-x_j/j}$ and the growth bases $\rho_j^{-1} = e^{x_j/j}$ now follow by ordinary exponential composition, with the error propagated explicitly. The theorem is a large- j statement about the pole locations; it does not by itself make the fixed- j coefficient estimate uniform when j grows with n .

[write] **Added 2 October 2026, batch 75: an instance of repository results.** The manuscript titled Theorem 7.3 “Pole accumulation transseries”; it is a single-scale Lambert- W expansion of the pole positions, and the theorem has been renamed accordingly. Taking logarithms, (7.7) reads $X + 2 \log X = \log(2j) - \log(1 + c/X)$. Its dominant block $X + 2 \log X = \log(2j)$ is solved exactly by the Lambert core `p0:thm:lambert-core` of the transseries-and-inversion volume [5] (there $a = 1$, $b = 2 > 0$, principal branch), which gives $w_j = 2W(\sqrt{j/2})$. The correction equation (7.12) is that volume’s reversion about an arbitrary core, `p0:thm:core-reversion`, with $\Lambda = 1$, $h = 0$ and small parameter $t = z = 1/w_j$; its triangular recurrence is the coefficient recursion described in the proof, and `code/expansion.py` implements it. The leading equation of Corollary 7.4, in the variable $x + c$, is another instance of that core (there $a = b = 1$), which gives $x + c = W(4/t)$. These are instances of that apparatus, and no novelty is claimed for the method. What is specific to this report is the pole-scale equation (Lemma 7.2) and the error terms in (7.9) and (7.10).

7.3 Inverting the real-pole counting function

Corollary 7.4. *Let $N_+(r) = \#\{j \geq 1 : \rho_j \leq r\}$, and put $t = -\log r$. Then, as $r \uparrow 1$,*

$$N_+(r) = \frac{W(4/t) - \log 2}{t} + O\left(\log \frac{1}{t}\right). \quad (7.14)$$

Proof. At $r = \rho_j$, rewrite (7.4) as

$$\frac{2e^{-x}}{t} = x + c + O(tx^2), \quad x = jt.$$

The leading equation has solution $x = W(4/t) - c$, since $e^c = 2$. Its derivative has order x , so the same two-stage mean-value argument gives an error $O(tx)$ in x , or $O(x)$ in j . Also $x \asymp \log(1/t)$. Between consecutive real poles, monotonicity sandwiches the counting function between the two adjacent values. The leading expression in (7.14) is decreasing in t , so the same error estimate holds throughout the gaps. $\square$

8 Inverting the area-counting asymptotics

8.1 Finite-sector real interpolants

Choose a cutoff $R > \rho$ as in Theorem 5.1. The finite principal-part expansion can be interpolated for real x as

$$F_R(x) = Ce^{Lx}(1 + E_R(x)), \quad L = \log \lambda > 0. \quad (8.1)$$

For a nonreal conjugate pair, use its real oscillatory expression; for a negative real pole use $r^{-x} \cos(\pi x)$ in place of $(-r)^{-n}$, with the same polynomial prefactor. This prescription agrees with every integer coefficient contribution and is real analytic. All nonleading terms and each fixed derivative are bounded by a polynomial times $e^{-\delta x}$, for some $\delta > 0$, because there are only finitely many poles in the chosen expansion. Consequently F_R is strictly increasing for sufficiently large x . This is a specified asymptotic interpolant, not a claim that a discrete sequence has a unique natural real interpolation.

Theorem 8.1 (Inverse-sector expansion). *Let $x_R(Y)$ solve $F_R(x) = Y$ on its eventually increasing branch, and put $x_0 = L^{-1} \log(Y/C)$. Set $\Phi(x) = -L^{-1} \log(1 + E_R(x))$. Then*

$$x_R(Y) = x_0 + \sum_{k \geq 1} \frac{1}{k!} \left(\frac{d}{dx} \right)^{k-1} \Phi(x)^k \Big|_{x=x_0}. \quad (8.2)$$

For each fixed cutoff and sufficiently large Y , this is a convergent small-correction expansion. In particular,

$$x_R(Y) = x_0 - \frac{E_R(x_0)}{L} + \frac{E_R(x_0)^2}{2L} + \frac{E_R(x_0)E'_R(x_0)}{L^2} + O(P(x_0)e^{-3\delta x_0}) \quad (8.3)$$

for a suitable fixed polynomial P .

Proof. Writing $x = x_0 + h$ and taking logarithms in (8.1) gives the exact equation $h = \Phi(x_0 + h)$. On a fixed small complex neighborhood of $h = 0$, Φ and its derivative tend uniformly to zero as $x_0 \rightarrow \infty$. Analytic Lagrange inversion applies; it may also be obtained by inserting a parameter s into $h = s\Phi(x_0 + h)$, expanding at $s = 0$, and using the uniform smallness to continue the series past $s = 1$. This proves (8.2). Expanding $\Phi = -E_R/L + E_R^2/(2L) + O(E_R^3)$, the terms with $k = 1, 2$ give (8.3); all remaining terms have at least three exponentially small factors, with polynomial derivative bounds. $\square$

For one simple real sector $E_R(x) = be^{-\Delta x}$, the first terms are especially transparent:

$$x_R(Y) = x_0 - \frac{b}{L} \left(\frac{Y}{C} \right)^{-\Delta/L} + \left(\frac{1}{2L} - \frac{\Delta}{L^2} \right) b^2 \left(\frac{Y}{C} \right)^{-2\Delta/L} + \cdots. \quad (8.4)$$

For the ρ_k sector, $b = C_k/C$ and $\Delta = \log(\rho_k/\rho)$. With several poles, products create mixed exponent sums. One must retain any other sector of comparable or larger size before interpreting a displayed correction as the next term of an ordered asymptotic expansion.

8.2 Integer thresholds and the rounding obstruction

Proposition 8.2. *The sequence $d(n)$ is strictly increasing for $n \geq 1$. Define $N(Y) = \min\{n \geq 1 : d(n) \geq Y\}$. For each fixed cutoff R , there exists $K_R > 0$ such that for all sufficiently large Y ,*

$$\lceil x_R(Y) - K_R Y^{-\eta_R} \rceil \leq N(Y) \leq \lceil x_R(Y) + K_R Y^{-\eta_R} \rceil, \quad \eta_R = \frac{\log(R/\rho)}{\log \lambda} > 0. \quad (8.5)$$

Proof. Append a part 1 to any valid composition. No new valley is created, since the new step is nonascending; this is an injection from area n to area $n + 1$. The singleton $(n + 1)$ is not in its image, proving strict increase.

At integers, Theorem 5.1 gives $d(n) = F_R(n) + O_R(R^{-n})$. Near $x_R(Y)$, $F'_R(x) \asymp Y$, so the corresponding uncertainty in the crossing location is bounded by a constant times $R^{-x_R}/Y = O(Y^{-\eta_R})$. One may make this precise first within a fixed distance of the crossing, using the mean value theorem; outside that distance the strict leading exponential separates the two sides. All integer points below the left shift have count below Y , and all those above the right shift have count at least Y . Taking ceilings yields the bounds. $\square$

A single unconditional identity $N(Y) = \lceil x_R(Y) \rceil$ would be unjustified when Y lies arbitrarily close to an actual integer threshold. Formula (8.5) records the missing rounding information rather than hiding it in a real-variable error term.

[write] **Added 2 October 2026, batch 75: an instance of repository results.** Proposition 8.2 and the remark after it are the staircase theorem `p0:thm:staircase` of the transseries-and-inversion volume [5]: part (1) gives $N(Y) = \lceil \nu(Y) \rceil$ for any increasing interpolation ν of the sequence, and part (2), the separation condition with $\eta = K_R Y^{-\eta_R}$, says that the two ceilings in (8.5) coincide, and then equal $N(Y)$, unless $x_R(Y)$ lies within η of an integer. The generic arithmetic of those two parts is formalized in Lean as `Fabius.staircase_ceil` and `Fabius.staircase_separation` in `Analysis/FabiusFunction/Lean/FabiusFunction/StaircaseInversion.lean`, statements about an arbitrary monotone interpolation that give no formal status to this proposition. The series (8.2) of Theorem 8.1 is Lagrange's reversion formula (in formal form, `p0:thm:LB` of the same volume). No novelty is claimed for these methods.

9 Reproducible computations and proof checks

9.1 Independent exact enumeration

The package includes two logically independent finite checks. First, a triangular coefficient algorithm implements (2.4)–(2.6) using integers and formal-series long division. Initializing $U_{N+1} = D_{N+1} = 0$ gives exactly all coefficients through degree N ; omitted objects have area larger than N . Second, exhaustive enumeration generates every composition through area 17, collapses consecutive equal parts, extracts interior local minima, and checks their order directly. It agrees with the coefficient algorithm. Lower bounds on the parts equal to 2 and 3 are also checked independently through area 13.

The nonempty U_1 coefficients agree with the displayed A001523 values through $n = 20$; its zero-degree coefficient differs only because $A = 1 + U_1$. The exact computation also gives

$$d(100) = 233832578565556052089825216845.$$

The program retains all coefficients through 200 in `verification.json`. Finite checks support the implementation; they are not substitutes for the asymptotic proofs.

9.2 Convergence of the finite ratios

Let $\varepsilon_n = d(n)/(C\lambda^n) - 1$. High-precision numerical comparison gives:

n	$d(n)/d(n-1)$	ε_n
10	2.00000000000000000000	$-6.7798174 \cdot 10^{-2}$
20	1.9808570502327331233	$-9.1891465 \cdot 10^{-3}$
30	1.9769260944011981878	$-1.0612096 \cdot 10^{-3}$
50	1.9764273761888468482	$-1.4002503 \cdot 10^{-5}$
75	1.9764207075861918763	$-6.1918466 \cdot 10^{-8}$
100	1.9764206780767492582	$-2.7346103 \cdot 10^{-10}$
150	1.9764206779458382883	$-5.3327913 \cdot 10^{-15}$
200	1.9764206779458357354	$-1.0399431 \cdot 10^{-19}$

The finite-area quantities reproduce both estimates in (1.1):

$$\frac{d(100)}{d(99)} = 1.976420678076749258223519681232\dots,$$

$$\frac{d(100)}{(d(100)/d(99))^{100}} = 0.603901514896651955795442836948\dots$$

This identifies the source of the discrepancy exactly; it is not a change in the combinatorial definition or in the area indexing.

9.3 Files and execution

`verify.py` requires only the Python standard library. It checks exact enumeration, agreement with the supplied classical initial terms, and the rigorous rational intervals. Run

```
python verify.py
```

The separate `numerics.py` requires `mpmath` and produces high-precision exploratory values for poles, residues, and limit-law constants. Its output explicitly distinguishes those numerical values from certified enclosures. A small optional `expansion.py` uses `sympy` to regenerate the $\alpha_k(c)$ coefficients from (7.12). The article is self-contained and can be compiled with

```
pdflatex article.tex
pdflatex article.tex
```

No source font files, downloaded papers, or repository files are bundled.

[write] **Added 2 October 2026, batch 75.** In the repository the programs live in `code/` and their recorded outputs in `data/`. Each program writes its output file into its own directory (`verify.py` writes `verification.json`, `numerics.py` writes `numerical_results.json` and imports `verify`, `expansion.py` writes `expansion_coefficients.json`), so the report's README runs them on a scratch copy. The delivered build script and PDF are not shipped; the README gives the build command.

10 Further research questions and topics

10.1 The nonreal pole spectrum

Determine the smallest-modulus nonreal poles, and certify their multiplicities and ordering relative to $\rho_2, \rho_3, \dots$. An effective route is to combine the explicit tail bounds with argument-principle or Rouché certificates on rationally described contours. This would turn the abstract

exponential cutoff in Theorem 5.1 into a numerically explicit first several sectors, and could yield sharp eventual error signs or ratio monotonicity. These latter conclusions have *not* been inferred from the numerical table.

10.2 Is the unit circle a natural boundary?

The real poles prove non-D-finiteness and obstruct meromorphic extension through 1. They do not prove a natural boundary at every point of the unit circle. A proof would need to locate noncancelling singularities approaching a dense set of boundary points, or otherwise preclude local continuation there. The factors Q_j and the root-of-unity structure of the unimodal products give a concrete starting point.

10.3 The next sector of pole accumulation

The estimate $x_j - X_j = O((\log j)^2/j)$ is beyond all logarithmic orders. Determine its leading coefficient and a full expansion in powers of $1/j$, with coefficient functions of w_j . The proof suggests a systematic calculation: retain the $k = 2$ term in (7.6), Euler–Maclaurin corrections to $(e^t - 1)^{-1}$, and a sharper weighted telescoping expansion in Lemma 7.1. The resulting two-scale transseries should improve the pole-counting error beyond $O(\log(1/t))$.

10.4 Joint limits and extreme heights

The renewal representation can mark the number of columns, the number of ones, peak heights, and the maximum height inside a repeated unit. A joint normal law for additive statistics should follow after proving nondegeneracy of the appropriate covariance matrix. Maximum height is not additive: determine whether its logarithmic scale exhibits a lattice Gumbel law and quantify any periodic phase. No extreme-value law is claimed in this report.

10.5 Moving lower bounds and a crossover problem

Theorem 4.1 holds for fixed j , while Theorem 7.3 describes large j . Determine a uniform estimate for $d_j(n)$ when $j = j(n)$ grows. Since the pole gap shrinks as j increases, a one-pole approximation need not remain uniform. Locating the scale at which multiple poles interact would connect the fixed-level exponential regime with sparse-composition regimes.

10.6 Formalization and effective algorithms

A useful ProveIt contribution would formalize the finite combinatorial decomposition first, including the plateau convention, then develop a general locally finite pole-product lemma. The exact interval certificate is suitable for a small independently checked arithmetic kernel. A separate algorithmic question is whether the exact coefficients can be obtained substantially faster than triangular long division, using the rapidly increasing minimum degrees of the kernels and fast truncated multiplication. Non-P-recursiveness rules out a fixed polynomial recurrence, not fast coefficient algorithms.

A A coefficient recurrence suitable for exact checking

Let all series be truncated modulo q^{N+1} . For descending $j = N, N-1, \dots, 1$, compute

$$\begin{aligned} Q_j &= 1 - q^j - q^j U_{j+1}, \\ (1 - q^j)^2 U_j &= q^j (1 - q^j) + U_{j+1}, \\ (1 - q^j) Q_j D_j &= q^j Q_j + D_{j+1}. \end{aligned} \tag{A.1}$$

Every denominator has constant term one. If $v(q)w(q) = b(q)$, $v(0) = 1$, its coefficients are obtained by

$$w_n = b_n - \sum_{k=1}^n v_k w_{n-k}.$$

This requires no division in the coefficient ring. Initialization with zero at level $N+1$ is exact at the requested order, so integrality of every computed coefficient is automatic. The enumeration proof in Section 2, not integrality alone, identifies the objects being counted.

B Claim ledger and limitations

Claim	Status and justification
Published generating function	Prior result of [1]; independently rederived by the unique minimum-part decomposition.
Pure exponential equivalent	Proved in Theorem 4.1, with exact constants and an exponentially smaller error.
Corrected numerical constants	Certified by standard-library integer interval arithmetic with explicit infinite-tail bounds.
Infinite positive poles; non-D-finiteness	Proved without a noncancellation assumption in Theorem 3.4 and Corollary 3.5.
Arbitrary exponential cutoff	Proved in Theorem 5.1; not a claim of convergence of an infinite pole sum.
Limit laws	Proved by uniform analytic perturbation and direct characteristic-function convergence.
All logarithmic pole orders	Proved in Theorem 7.3; the $1/j$ -scale correction remains to be sharpened.
Inverse expansion	Proved for a specified finite-sector interpolant; integer crossings are bounded with two ceilings.
Novelty and formal status	No matching prior proof found in the targeted search. No exhaustive priority claim, no Lean proof, and no independent peer review are asserted.

References

- [1] R. Flórez, J. L. Ramírez, and D. Villamizar, *Restricted bargraphs and unimodal compositions*, Journal of Combinatorial Theory, Series A **208** (2024), 105934. [doi:10.1016/j.jcta.2024.105934](https://doi.org/10.1016/j.jcta.2024.105934). Especially Section 2 and Conjecture 2.3. Publisher HTML consulted October 1, 2026.

- [2] OEIS Foundation Inc., *A001523: Number of stacks, or planar partitions of n ; also number of weakly unimodal compositions of n* . <https://oeis.org/A001523>. Entry consulted October 1, 2026. The entry credits Auluck's known leading asymptotic and links the bargraph paper above.
- [3] P. Flajolet and R. Sedgewick, *Analytic Combinatorics*, Cambridge University Press, 2009. Chapters IV–V for rational and meromorphic asymptotics, and Chapter IX for parameter distributions. Author's book page:
<https://sedgewick.io/books/analytic-combinatorics/>.
- [4] V. Reshetnikov, *ProveIt*, project repository, particularly `Analysis/Transseries/README.md`.
<https://github.com/VladimirReshetnikov/ProveIt>. Read-only inspection, October 1, 2026. No repository changes were made.
- [5] *Transseries and inversion*, a volume of the ProveIt repository:
`Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/transseries_and_inversion.tex` (theorems `p0:thm:lambert-core`, `p0:thm:core-reversion`, `p0:thm:staircase` and `p0:thm:LB`). Added in writing, batch 75.